

Number Sense

Marking Scheme & Blueprint

MTH1W · Module 1 Test — Teacher Copy

Blueprint

Total: 35 marks · 50 minutes. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	Integers	L1	2	K/U	A
A2	Order of Operations	L1	2	K/U	A
A3	Exponent Rules	L1	2	K/U	A
A4	Fractions	L1	2	K/U	A
A5	Fractions ↔ Decimals ↔ Percent	L1	2	K/U	A
A6	Square Roots & Irrationals	L1	2	K/U	A
A7	Applying Percents	L1	2	K/U	A
A8	Ratios, Rates & Proportions	L1	2	K/U	A
B9	Fractions	L2	3	App	B
B10	Applying Percents	L2	3	App	B
B11	Exponent Rules	L2	2	App	B
B12	Ratios, Rates & Proportions	L2	3	App	B
C13	Applying Percents (multi-step)	L3	5	Think	C
D14	Square Roots & Irrationals	L3	3	Comm	D

By category (marks): K/U 16 · Application 11 · Thinking 5 · Communication 3

By level (marks): L1 16 · L2 11 · L3 8 → L1 45% · L2 31% · L3 22%

Worked Solutions & Mark Allocation

1. (A1 · K/U · 2 marks)

$(-15) + (-6) + (+4) = (-21) + 4 = -17$. (A1 setup, A1 answer)

2. (A2 · K/U · 2 marks)

$2^3 = 8$; $4 \times 8 = 32$; $32 \div 8 = 4$; $30 - 4 = 26$. (M1 order, A1 answer)

3. (A3 · K/U · 2 marks)

Add then subtract exponents: $6^{5+2-3} = 6^4$. (M1 method, A1 answer)

4. (A4 · K/U · 2 marks)

$\frac{6}{8} - \frac{5}{8} = \frac{1}{8}$. (M1 common denom, A1 answer)

5. (A5 · K/U · 2 marks)

$5 \div 8 = 0.625 = 62.5\%$. (A1 decimal, A1 percent)

6. (A6 • K/U • 2 marks)

44 is not a perfect square $\rightarrow$ irrational. $6^2=36 < 44 < 49=7^2$, so between 6 and 7.
(A1 classify, A1 bounds)

7. (A7 • K/U • 2 marks)

$0.30 \times 250 = 75$. (A2 answer)

8. (A8 • K/U • 2 marks)

$180 \div 3 = 60$ km/h. (M1 setup, A1 answer)

9. (B9 • Application • 3 marks)

$\frac{2}{3} + \frac{1}{2} = \frac{4}{6} + \frac{3}{6} = \frac{7}{6}$. (M1)

$\frac{7}{6} \times \frac{6}{5} = \frac{42}{30} = \frac{7}{5}$. (M1 multiply, A1 answer)

10. (B10 • Application • 3 marks)

Total = $540 \times 1.13 = \$610.20$. (M1 method, A1 total)

Tax = $610.20 - 540 = \$70.20$. (A1 tax)

11. (B11 • Application • 2 marks)

$(5^2)^2 = 5^4$; $5^4 \div 5^3 \times 1 = 5^1 = 5$. (M1 laws, A1 value)

12. (B12 • Application • 3 marks)

Total parts = $3 + 5 = 8$; one part = $4000 \div 8 = 500$ mL. (M1)

Orange = $3 \times 500 = 1500$ mL; pineapple = $5 \times 500 = 2500$ mL. (A1 orange, A1 pineapple)

13. (C13 • Thinking • 5 marks)

Let p be the original price. After +20% then -20%: $p \times 1.20 \times 0.80 = p \times 0.96$. (M1 model)

$p \times 0.96 = 96 \rightarrow p = 96 \div 0.96 = \100 . (M1 solve, A1 answer)

Final = 96, original = 100, so they are NOT equal. (A1)

Reason: +20% then -20% multiplies by 0.96, i.e. 96% of the original — a net 4% loss. The percentages act on different amounts. (A1 reasoning)

14. (D14 • Communication • 3 marks)

$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$, but $\sqrt{25} = 5$. (A1 calculation)

Since $7 \neq 5$, Jordan is not correct. (A1 conclusion)

Explanation: the square root does not distribute over addition; in general $\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$. (C1 communication)